

Alzheimer's Severity Classifier

*An Interpretable Biomarker-Based Machine Learning Pipeline
for MRI Brain Scan Classification*

Technical Project Report

Multi-Layer Perceptron | 44,000 MRI Scans | 7 Biomarkers | 4 Severity Classes

Abstract. This report describes a complete machine learning pipeline for classifying Alzheimer's disease severity from grayscale brain MRI images. Rather than using an end-to-end convolutional neural network on raw pixels, the system extracts seven clinically interpretable biomarkers from each scan — including brain area, ventricle ratio, grey matter percentage, white matter percentage, CSF percentage, mean intensity, and texture — and feeds these into a trained Multi-Layer Perceptron (MLP) classifier. Four models were evaluated: MLP, Random Forest, XGBoost, and Linear SVM. The tuned MLP achieves the best performance with 62.8% cross-validation accuracy and 62.3% test accuracy on a 44,000-image dataset with four severity classes. A real-time Streamlit web application delivers predictions with confidence scores and biomarker breakdowns.

1. Introduction

Alzheimer's disease is the most prevalent form of dementia, affecting over 55 million people globally. Accurate staging of disease severity from neuroimaging data is of clinical importance for treatment planning, patient monitoring, and clinical trial enrolment. MRI brain scans offer a non-invasive window into structural brain changes — including tissue atrophy, ventricular enlargement, and alterations in grey and white matter composition — that correlate with disease progression.

This project builds an interpretable, end-to-end classification pipeline that: (1) processes raw MRI images through a reproducible preprocessing pipeline; (2) extracts seven scalar biomarkers capturing structural and intensity properties of the brain; and (3) classifies each scan into one of four severity categories using a trained machine learning model. The system is deployed as an interactive Streamlit web application.

1.1 Design Philosophy

The central design decision of this project is to use hand-crafted biomarker features rather than training a convolutional neural network (CNN) directly on pixel data. This choice is motivated by three priorities:

- **Interpretability:** Every prediction is grounded in named, clinically meaningful biomarker values that can be read, verified, and questioned.
- **Computational efficiency:** Extracting 7 features and running a small MLP takes milliseconds on any CPU, with no GPU infrastructure required.
- **Robustness:** The normalisation pipeline removes scanner-dependent intensity offsets, improving cross-scanner generalisation.

1.2 Severity Classification Schema

The system classifies each MRI scan into one of four severity levels:

Class	Label	Description
0	Non Demented	No detectable cognitive impairment; normal brain structure.
1	Very Mild Demented	Earliest detectable signs; minimal structural change.
2	Mild Demented	Moderate cognitive decline with visible structural changes.
3	Moderate Demented	Significant tissue loss, enlarged ventricles, marked atrophy.

Table 1. Alzheimer's severity classification schema.

2. Dataset

The dataset comprises 44,000 grayscale brain MRI images spanning all four severity classes. Images are sourced from a publicly available Alzheimer's MRI dataset and represent a variety of acquisition conditions and patient demographics. The dataset is partitioned into an 80% training set (35,200 images) and a 20% held-out test set (8,800 images).

2.1 Data Splits and Cross-Validation

To ensure class balance across evaluation folds, 10-Fold Stratified Cross-Validation is applied during model training. Stratification guarantees that each fold contains the same proportion of each severity class as the full dataset, preventing biased evaluation arising from class imbalance in any individual fold.

Split	Size	Purpose
Training set	35,200	10-fold stratified CV for model training and selection
Test set	8,800	Final held-out evaluation; not seen during training or tuning
Total	44,000	Full dataset across all four severity classes

Table 2. Dataset split statistics.

2.2 Pre-Extracted Biomarker Files

Biomarkers are extracted once from all 44,000 images and stored as CSV files to accelerate experimentation and ensure reproducibility:

- **full_train_biomarkers.csv** — 35,200 rows; training set biomarker features and labels.
- **full_test_biomarkers.csv** — 8,800 rows; test set biomarker features and labels.
- **phase1_biomarkers.csv** — Intermediate Phase 1 extraction output.

Each CSV contains the 7 biomarker columns alongside a string class label and an integer severity label (0–3), ready for direct ingestion by scikit-learn.

3. Image Preprocessing Pipeline

All MRI images are passed through a deterministic four-step preprocessing pipeline before biomarker extraction. This pipeline ensures spatial consistency and intensity comparability across all scans regardless of original image dimensions or scanner settings.

3.1 Step 1 — Crop to Brain

Background pixels with intensity at or below 10 are identified. The bounding box of foreground brain pixels is computed along both axes, and the image is cropped to this region with a 10-pixel margin retained on each side to avoid clipping brain edges.

```
crop_to_brain(arr, margin=10)
```

3.2 Step 2 — Pad to Square

The cropped image — which may be rectangular — is zero-padded on the shorter axis to produce a square canvas. The brain region is centred within the canvas.

```
pad_to_square(arr)
```

3.3 Step 3 — Resize to 224 x 224

The square image is resized to a fixed 224 x 224 pixel grid using bicubic interpolation. This standard resolution ensures all subsequent biomarker computations operate on the same spatial scale.

```
Image.resize((224, 224), Image.BICUBIC)
```

3.4 Step 4 — Min-Max Normalisation

Brain pixel intensities are normalised to the range [0, 1] using the minimum and maximum values of foreground (non-background) pixels. Background pixels are set to 0. This removes scanner-dependent intensity offsets and puts all scans on a common scale.

```
normalize_minmax(arr) # background pixels forced to 0.0
```

4. Biomarker Extraction

Following preprocessing, seven scalar biomarkers are extracted from each normalised image. These biomarkers form the feature vector passed to the machine learning classifier.

4.1 Brain Mask Generation

A binary brain mask is generated using Otsu's thresholding — an automatic method that finds the optimal intensity cutoff to minimise intra-class variance between brain tissue and background. Small connected components with fewer than 500 pixels are removed using morphological filtering, and holes within the brain region are filled using binary fill holes, producing a clean, solid brain mask.

4.2 Tissue Segmentation

Within the brain mask, pixels are classified into tissue types based on normalised intensity ranges. These thresholds are calibrated to the min-max normalised [0, 1] scale:

Tissue Type	Intensity Range	Description
Cerebrospinal Fluid (CSF)	0.01 – 0.25	Dark fluid-filled spaces within the brain
Grey Matter	0.25 – 0.65	Intermediate intensity; neuron cell bodies
White Matter	≥ 0.65	Bright regions; myelinated axon fibres

Table 3. Intensity-based tissue segmentation thresholds.

4.3 Ventricle Detection

Ventricle regions are detected using a percentile-based approach. Within the brain mask, pixels below the 20th percentile of brain intensity are classified as ventricles. This approach adapts automatically to each scan's intensity distribution. Small objects with fewer than 100 pixels are subsequently removed to exclude noise.

4.4 The Seven Biomarkers

The following seven biomarkers are extracted from each scan:

Feature	Computation	Clinical Relevance
brain_area	Total pixel count within brain mask	Proxy for brain volume; decreases with atrophy
ventricle_ratio	Ventricle pixels / mask pixels x 100	Enlarged ventricles indicate neurodegeneration
csf_pct	CSF pixels / mask pixels x 100	Increasing CSF indicates tissue loss
grey_pct	Grey matter pixels / mask pixels x 100	Grey matter loss is a hallmark of Alzheimer's
white_pct	White matter pixels / mask pixels x 100	White matter integrity assessment
mean_intensity	Mean intensity of all brain mask pixels	Overall tissue density
std_intensity	Std deviation of brain mask intensities	Textural heterogeneity of brain tissue

Table 4. The seven extracted biomarker features.

5. Models and Evaluation

Four machine learning models were trained and evaluated on the same seven-biomarker feature set using identical train/test splits and 10-Fold Stratified Cross-Validation.

5.1 Models Trained

- **MLP (Neural Net):** Multi-Layer Perceptron from scikit-learn. Multiple hidden layers allow the model to learn non-linear decision boundaries in the 7-dimensional biomarker space. Both a baseline and a hyperparameter-tuned version were trained.
- **Random Forest:** Ensemble of decision trees with bootstrap aggregation. Provides built-in feature importance scores and is robust to outliers.
- **XGBoost:** Gradient-boosted tree ensemble. Sequential trees correct the residual errors of previous trees, offering strong tabular data performance.
- **SVM (Linear):** Support Vector Machine with a linear kernel. Tests whether the biomarker feature space is linearly separable.

5.2 Evaluation Metrics

Each model is evaluated on three primary dimensions: (1) cross-validation accuracy averaged across 10 folds with standard deviation; (2) held-out test accuracy; and (3) the overfitting gap, computed as CV accuracy minus test accuracy. A gap below 0.10 is considered indicative of a well-generalised model. F1 score and precision are also reported.

5.3 Results

Model	CV Acc	Std	Test Acc	CV F1	Test F1	Gap
MLP (Neural Net)	0.628	0.009	0.623	0.624	0.623	0.005
Random Forest	0.594	0.006	0.583	0.587	0.575	0.011
XGBoost	0.543	0.007	0.528	0.533	0.518	0.015
SVM (Linear)	0.389	0.004	0.381	0.340	0.333	0.007

Table 5. Model performance comparison. All gaps are below the 0.10 overfitting threshold.

5.4 Key Findings

- The tuned **MLP (Neural Net)** achieves the highest performance on all metrics — 62.8% CV accuracy and 62.3% test accuracy — with the smallest overfitting gap of 0.005.
- All four models remain well below the 0.10 overfitting threshold, demonstrating that seven biomarkers form a stable, non-overfit feature representation.
- **SVM (Linear)** performs worst at 38.9% CV accuracy, confirming that the decision boundary in biomarker space is fundamentally non-linear.
- **XGBoost** exhibits the highest overfitting gap (0.015), suggesting boosting begins to exploit noise in the training biomarkers.

- The near-identical CV and test F1 scores for MLP (0.624 vs 0.623) confirm the model does not overfit to any particular fold or class distribution.

5.5 Accuracy in Context

Achieving approximately 62% accuracy on a four-class MRI classification task using only seven hand-crafted biomarkers is a strong result. The theoretical random-chance baseline for four balanced classes is 25%. End-to-end CNN models trained directly on raw pixels can achieve substantially higher accuracy (typically above 90%), but at the cost of complete opacity — the model cannot explain the basis for any individual prediction. This project deliberately prioritises interpretability over raw performance.

6. Streamlit Web Application

The trained MLP model is deployed as a real-time interactive web application using the Streamlit framework. The application accepts a brain MRI image upload and returns a complete prediction result within seconds, with no server-side GPU requirements.

6.1 Application Inference Flow

- User uploads a brain MRI image (JPG, JPEG, or PNG) via the file uploader widget.
- The image is preprocessed: grayscale conversion, crop, pad, resize to 224x224, min-max normalisation.
- Seven biomarkers are extracted via Otsu thresholding, morphological operations, and intensity-range tissue segmentation.
- The 7-feature vector is passed to the loaded MLP model (model_mlp_tuned.pkl).
- The model returns a severity class index and full four-class probability distribution.
- Results are rendered across four UI panels.

6.2 UI Panels

Panel	Content
MRI Display	Uploaded scan image rendered at full quality.
Prediction Panel	Severity label, confidence score with progress bar, and full four-class probability distribution with colour-coded bars.
Biomarker Panel	Table of all seven extracted biomarker values with descriptive labels.
Feature Importance	Horizontal bar chart of model feature importances, where available.
Sidebar	Pipeline description, severity class legend, model status indicator, and dataset statistics.

Table 6. Application UI panel descriptions.

6.3 Model Loading

The model is loaded once at application startup and cached using Streamlit's resource cache:

```
import joblib
model = joblib.load('model_mlp_tuned.pkl')
X = np.array([[biomarkers[f] for f in BIOMARKER_FEATURES]])
severity_idx = int(model.predict(X)[0])
probas = model.predict_proba(X)[0]
```

7. Project Structure

File	Type	Description
app.py	Python	Streamlit web application; full inference pipeline.
alz.ipynb	Notebook	Training, evaluation, and visualisation notebook.
requirements.txt	Config	Pinned Python package dependencies.
model_mlp_tuned.pkl	Model	Deployed tuned MLP classifier (best performance).
model_mlp.pkl	Model	Baseline untuned MLP classifier.
model_svm.pkl	Model	Trained SVM (Linear) classifier.
model_bst.pkl	Model	Trained boosted tree model.
xgb_alzheimer_model.pkl	Model	Trained XGBoost classifier.
full_train_biomarkers.csv	Data	35,200-row training biomarker features and labels.
full_test_biomarkers.csv	Data	8,800-row test biomarker features and labels.
train_split.csv	Data	Raw training split image paths and indices.
test_split.csv	Data	Raw test split image paths and indices.
phase1_biomarkers.csv	Data	Phase 1 intermediate biomarker extraction output.
model_comparison.csv	Results	Metrics table for all four models.
all_model_results.csv	Results	Detailed per-model evaluation results.
model_comparison.png	Figure	Bar chart comparing CV and test accuracy for all models.

Table 7. Project file directory.

8. Installation and Usage

8.1 Prerequisites

- Python 3.9 or higher
- pip package manager
- model_mlp_tuned.pkl in the same directory as app.py

8.2 Setup

```
git clone https://github.com/YOUR_USERNAME/alzheimers-classifier.git cd
alzheimers-classifier pip install -r requirements.txt streamlit run app.py
```

The application opens at <http://localhost:8501> in the default browser.

8.3 Dependencies

Package	Version	Purpose
streamlit	latest	Web application framework
scikit-learn	1.6.1	MLP, SVM classifiers and evaluation utilities
scikit-image	0.25.2	Otsu thresholding and morphological operations
scipy	1.15.2	Binary fill holes for mask generation
pillow	10.4.0	Image loading and bicubic resizing
pandas	2.2.3	CSV data loading and handling
numpy	2.2.3	Array and matrix operations
joblib	1.4.2	Model serialisation and .pkl file loading

Table 8. Python package dependencies.

9. Technical Notes

9.1 Feature Vector Order

The model expects input features in a fixed order during inference. The application enforces this via the BIOMARKER_FEATURES constant in app.py:

```
BIOMARKER_FEATURES = ['std_intensity', 'brain_area', 'grey_pct',
                       'ventricle_ratio', 'white_pct', 'mean_intensity', 'csf_pct']
```

This order must match exactly the column order used during model training in alz.ipynb.

9.2 Otsu Thresholding

Otsu's method computes the optimal intensity threshold that minimises intra-class intensity variance, making brain-background segmentation robust across different scanners and acquisition protocols without requiring manual threshold tuning.

9.3 Hyperparameter Tuning

The tuned MLP (model_mlp_tuned.pkl) was optimised over hidden layer sizes, activation functions, learning rate schedules, and regularisation strength using grid or random search cross-validation. The tuned model outperforms the baseline MLP across all evaluation metrics.

10. Limitations and Future Work

10.1 Current Limitations

- **Accuracy ceiling:** Seven biomarkers limit the model's representational capacity. Raw-pixel CNNs can capture finer spatial patterns and achieve higher accuracy.
- **Fixed intensity thresholds:** The CSF, grey matter, and white matter intensity ranges (0.25 / 0.65) are fixed and may not generalise perfectly across all scanner types.
- **2D slices only:** The system processes individual 2D MRI slices rather than full 3D volumetric scans, losing inter-slice spatial context.
- **Not clinically validated:** The model has not been validated against expert radiologist labels or clinical diagnostic outcomes.

10.2 Future Directions

- **3D volumetric features:** Extend biomarker extraction to 3D MRI volumes for richer spatial information including cortical thickness and hippocampal volume.
- **Hybrid architecture:** Combine CNN spatial features with interpretable biomarkers for higher accuracy while preserving partial interpretability.
- **SHAP explanations:** Integrate SHAP values into the application to display each biomarker's contribution to individual predictions.
- **Multi-scanner validation:** Validate the pipeline on MRI data from multiple scanner manufacturers to assess cross-scanner robustness.
- **Longitudinal tracking:** Extend the application to monitor a patient's biomarkers over multiple scans to track disease progression.

Disclaimer: This project is produced for educational and research purposes only. It is not a clinically validated diagnostic tool and must not be used as a substitute for professional medical assessment by a qualified neurologist or radiologist.